

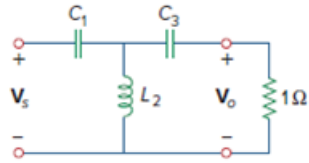
Problem

Synthesize the transfer function

$$H(s) = \frac{V_o}{V_s} = \frac{s^3}{s^3 + 6s + 12s + 24}$$

using the LC ladder network in Fig. 19.133.

Figure 19.133



Step-by-step solution

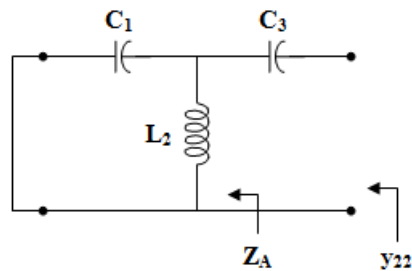
Step 1 of 1

$$H(s) = \frac{s^3}{(s^3 + 12s) + (6s^2 + 24)} = \frac{\frac{s^3}{s^3 + 12s}}{1 + \frac{6s^2 + 24}{s^3 + 12s}}$$

Hence,

$$y_{22} = \frac{6s^2 + 24}{s^3 + 12s} = \frac{1}{sC_3} + Z_A \quad (1)$$

where Z_A is shown in the figure below.



We now obtain C_3 and Z_A using partial fraction expansion.

$$\text{Let } \frac{6s^2 + 24}{s(s^2 + 12)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 12}$$

$$6s^2 + 24 = A(s^2 + 12) + Bs^2 + Cs$$

Equating coefficients :

$$s^0: 24 = 12A \longrightarrow A = 2$$

$$s^1: 0 = C$$

$$s^2: 6 = A + B \longrightarrow B = 4$$

Thus,

$$\frac{6s^2 + 24}{s(s^2 + 12)} = \frac{2}{s} + \frac{4s}{s^2 + 12} \quad (2)$$

Comparing (1) and (2),

$$C_3 = \frac{1}{A} = \frac{1}{2} \text{ F}$$

$$\frac{1}{Z_A} = \frac{s^2 + 12}{4s} = \frac{1}{4}s + \frac{3}{s} \quad (3)$$

$$\text{But } \frac{1}{Z_A} = sC_1 + \frac{1}{sL_2} \quad (4)$$

Comparing (3) and (4),

$$C_1 = \frac{1}{4} \text{ F} \quad \text{and} \quad L_2 = \frac{1}{3} \text{ H}$$

Therefore,

$$C_1 = \underline{0.25 \text{ F}}, \quad L_2 = \underline{0.3333 \text{ H}}, \quad C_3 = \underline{0.5 \text{ F}}$$